

# Problem Setup

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## Problem Setup

We are given the quadratic equation:

$$x^2 + 2x + 1 = 0$$

The objective is to find all values of  $x$  that satisfy this equation.

## Strategy and Key Principles

To solve the quadratic equation  $x^2 + 2x + 1 = 0$ , we can recognize the left-hand side as a standard algebraic identity, specifically a perfect square trinomial of the form  $a^2 + 2ab + b^2 = (a + b)^2$ . Alternatively, we can use the quadratic formula or factoring by grouping. Factoring using the perfect square identity provides the most direct and insightful derivation.

## Detailed Solution

We begin with the given quadratic equation:

$$x^2 + 2x + 1 = 0$$

Next, we examine the left-hand side of the equation to check if it can be factored. We observe that the first term is  $x^2$  (which is  $a^2$ ), the last term is 1 (which is  $b^2 = 1^2$ ), and the middle term is  $2x$  (which corresponds to  $2ab = 2(x)(1) = 2x$ ). Therefore, the expression is a perfect square trinomial.

We rewrite the left-hand side as a squared binomial:

$$(x + 1)^2 = 0$$

To solve for  $x$ , we take the square root of both sides of the equation:

$$\begin{aligned}\sqrt{(x + 1)^2} &= \sqrt{0} \\ x + 1 &= 0\end{aligned}$$

Finally, we isolate  $x$  by subtracting 1 from both sides:

$$x = -1$$

## Final Answer

$$\boxed{x = -1}$$

## Key Insights

- **Discriminant Analysis:** The discriminant  $\Delta = b^2 - 4ac$  for this equation is  $2^2 - 4(1)(1) = 4 - 4 = 0$ . A discriminant of zero indicates that the quadratic equation has exactly one distinct real root (or a repeated root of multiplicity 2).
- **Multiplicity:** The solution  $x = -1$  is a double root, meaning the factor  $(x + 1)$  appears twice in the fully factored form  $(x + 1)(x + 1) = 0$ .
- **Graphical Interpretation:** The graph of  $y = x^2 + 2x + 1$  is a parabola that opens upward with its vertex resting precisely on the  $x$ -axis at  $(-1, 0)$ , which explains why it has only one  $x$ -intercept.